

Validity Analysis: INJONGO

Target context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Overall risk: **HIGH**

Deployment Context

Use case: Amharic intercity-travel booking bots for Sky Bus, Selam Bus, and Ride covering routes, fares, and seat changes

Target population: Inter-regional commuters and travelers booking long-distance buses and ride-hail trips via smartphone apps

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology $\triangle$	2	Significant gaps	high
Input Content $\triangle$	3	Moderate gaps	high
Input Form $\checkmark$	5	Strong alignment	high
Output Ontology	2	Significant gaps	high
Output Content $\triangle$	2	Significant gaps	high
Output Form $\checkmark$	4	Minor gaps	high
Average	3.0		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

INJONGO is the most relevant existing benchmark for Amharic intent detection and slot-filling, and the only large-scale resource of its kind. For the Ethiopian intercity/ride-hail deployment, it provides genuine value in three respects: (1) Amharic is included as a first-class language with culturally grounded entities (Ethiopian banks, TeleBirr, Ethiopian cities including Dire Dawa, Bahir Dar, Hawassa, Arba Minch, and even Ge'ez calendar months Meskerem and Hamle as DATE slot values); (2) text/Ge'ez script and structured intent+slot output exactly match deployment requirements; (3) annotation quality after review is high ($\kappa=0.933$ for Amharic). However, three substantive gaps limit construct validity for this specific deployment: (a) the 40-intent travel taxonomy is air-travel-framed and entirely lacks intercity bus, seat-class, fare-tier, and operator-specific rebooking intents — confirmed empirically in the Amharic split (HIGH-priority IO concern); (b) Ethiopian Birr is absent as a

CURRENCY/MONEY slot value in the sampled Amharic data while other tracks' currencies (Naira, Rand) are well-represented, and most Amharic DATE slots use relative expressions rather than named Ge'ez months (HIGH-priority IC residuals); (c) no code-switched Amharic–Oromo / Amharic–Tigrinya utterances exist despite the L2-heavy multi-regional user base, and annotator regional demographics are undisclosed (HIGH-priority OC concern). The benchmark is suitable as a foundation but requires deployment-specific supplementation before being treated as sufficient evaluation evidence.

Practical Guidance

What This Benchmark Measures

INJONGO measures Amharic NLU performance on a 40-intent / 23-slot taxonomy spanning banking, home, kitchen-and-dining, travel, and utility domains, using accuracy (intent) and F1 (slot). It provides a valid measure of (i) general Amharic text comprehension by encoder, encoder-decoder, and LLM-prompted models, (ii) culturally grounded slot extraction for Ethiopian banks, payment platforms (TeleBirr), and major intercity cities, and (iii) cross-lingual transfer behavior from English to Amharic. Strongest dimensions for this deployment are input form (full text/Ge'ez match) and output form (structured intent+slot output matches NLU module). It can support model selection across encoder, decoder, and LLM families for the Amharic NLU layer.

Construct Depth

Construct depth is shallow-to-moderate for the deployment's actual task. The benchmark probes generic conversational NLU on Amharic and partially covers Ethiopian cultural grounding in banking and selected travel slots, but it does not probe (a) intercity bus operator booking flows, (b) seat-class/fare-tier reasoning, (c) Ethiopian Birr fare extraction in travel context, (d) Ethiopian-clock-convention TIME extraction, (e) Ge'ez-calendar dense temporal reasoning beyond the few sampled instances, or (f) code-switched Amharic–Oromo / Amharic–Tigrinya robustness. High benchmark scores would not justify confidence that a deployed model handles fare extraction, holiday-surge bookings, or L2-speaker code-switched utterances correctly.

What Else You Need

(1) For input ontology: build an operator-specific intent inventory (Sky Bus, Selam Bus, Ride) covering seat-class upgrade, road-closure rebooking, holiday-surge booking, and operator-specific cancellation; benchmark on this. (2) For input content: collect or synthesize an Amharic test slice with Ethiopian Birr-denominated fares, dense Ge'ez calendar month references including Pagume, Ethiopian-clock-convention TIME values, and the full set of intercity corridor toponyms; supplement with code-switched Amharic–Oromo and Amharic–Tigrinya utterances. (3) For output content: re-annotate or expert-validate a slice using non-Addis-centric L2 Amharic annotators, including rural and Tigray/Oromia-region speakers, to assess label validity for code-switched utterances. (4) Supplement with EthioBenchmark NER/POS as auxiliary evidence, and operator-log-derived utterances for ecological validity.

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

INJONGO's travel domain is derived from CLINC's generic intent set ('exchange rate', 'book flight') [\[Q39\]](#) and explicitly omits intercity bus operator flows. Empirical inspection of the Amharic split confirms that all sampled travel-domain Amharic utterances use air-travel framing (book_flight to Turkey, Sudan) or generic car rental — there are no intercity bus, seat-class, or operator-specific intents [\[DATASET-D7, D8, D25, D26\]](#). The deployment requires intent detection for Sky Bus / Selam Bus / Ride bus and ride-hail booking, including seat-class upgrade, road-closure rebooking, holiday-surge booking, and operator-specific cancellation, none of which appear in the 40-intent taxonomy. The paper itself acknowledges scope limits [\[Q107, Q111\]](#). Although elicitation marked IO as MODERATE priority (user accepts general travel intents as broadly comparable), the empirical ontology mismatch in the travel subdomain is concrete and substantial.

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g. " (p.3)
Q107	"The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications." (p.9)
Q111	"Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations." (p.9)
D7	ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንግረኝ። / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D25	intent=book_flight, COUNTRY: ቱርክ (Turkey) — Flight to Turkey — no intercity bus, seat class, or operator-specific slots present
D26	intent=book_flight, COUNTRY: ሱዳን (Sudan) — Flight to Sudan — air travel framing, not surface transport

Strengths

- Joint intent + slot taxonomy is well-defined and matches the deployment's NLU module structure [\[Q2, Q31\]](#)
- Travel is one of five included domains, providing partial overlap with bus booking flows [\[Q5, Q117\]](#)
- Ground-up African-context elicitation rather than translation reduces some Western framing [\[Q8, Q30\]](#)

ID	Evidence
Q2	"Intent detection and slot-filling are crucial components of the natural language understanding module in task-oriented dialogue systems (Hemphill et al., 1990; Coucke et al., 2018; Gupta et al., 2018)." (p.1)
Q5	"We cover the following five domains: banking, home, travel, utility, and kitchen & dining." (p.1)
Q8	"Through extensive experiments, we benchmark the fine-tuning multilingual transformer models and the prompting large language models (LLMs), and show the advantage of leveraging African-cultural utterances over Western-centric utterances for improving cross-lingual transfer from the English language." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)

Q31	"INJONGO is a joint intent detection and slot-filling dataset (ID-SF) for typologically diverse Sub-Saharan African languages and English." (p.2)
Q117	"These are a total of 40 categories across 5 domains (Banking, Kitchen and Dining, Travel, Utility, and Home)." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include intercity bus route queries, fare lookup, seat selection (VIP/economy), booking, cancellation/rebooking — only generic 'book_flight', 'car_rental', and 'travel_suggestion' overlap, established by elicitation and observed in the Amharic data [DATASET-D7, D8].

IO-2: Yes. INJONGO covers 5 domains and 40 intents derived from CLINC [Q39, Q117] and omits intercity bus, seat-class selection, holiday-surge booking, and road-closure rebooking. The paper explicitly notes the taxonomy 'missing some other domains such as healthcare and education' [Q107] and a non-natural intent frequency distribution [Q111].

IO-3: The travel domain is dominated by air travel framing in the Amharic split (international flights to Turkey, Sudan) [DATASET-D7, D8, D25, D26]; these are not necessarily irrelevant but concentrate evaluation weight on air rather than surface intercity transport relevant to the deployment.

IO-4: Documented gaps: (a) no intercity bus operator intents, (b) no seat-class/fare-tier intents, (c) no rebooking-on-disruption intent, (d) no Ethiopian-holiday surge booking intents — all of which harm content validity for this specific deployment.

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g." (p.3)
Q107	"The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications." (p.9)
Q111	"Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations." (p.9)
Q117	"These are a total of 40 categories across 5 domains (Banking, Kitchen and Dining, Travel, Utility, and Home)." (p.13)
D7	ወደ ቱርክ በፍጥነት የምደርስበት የበረራ አማራጭ ንግረኝ። / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D25	intent=book_flight, COUNTRY: ቱርክ (Turkey) — Flight to Turkey — no intercity bus, seat class, or operator-specific slots present
D26	intent=book_flight, COUNTRY: ሱዳን (Sudan) — Flight to Sudan — air travel framing, not surface transport

Information Gaps

- Whether any Amharic travel-domain utterances in the unsampled 99.4% of the dataset use surface-transport framing

Requires Expert Verification

- Definitive intent inventory of Sky Bus / Selam Bus / Ride booking APIs to quantify gap
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Remediation

Gap 1: INJONGO's 40-intent travel taxonomy is air-travel framed and lacks intercity bus operator flows (seat-class, rebooking, holiday-surge)

Recommendation: Define a deployment-specific intent supplement (~10–15 intents) drawn from Sky Bus / Selam Bus / Ride API surfaces; collect 50–100 Amharic utterances per intent via native speaker elicitation following INJONGO's methodology; treat this as a required held-out evaluation alongside INJONGO's travel domain.

Input Content (IC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Empirical dataset inspection partially resolves what the paper does not document. Ethiopian calendar months (Meskerem, Hamle) appear as DATE slot values [\[DATASET-D1, D2, D6\]](#), Ethiopian intercity cities (Dire Dawa, Bahir Dar, Hawassa, Arba Minch) appear as CITY_OR_PROVINCE slots [\[DATASET-D2, D9\]](#), and Ethiopia-specific financial entities (Amhara Bank, Commercial Bank of Ethiopia, TeleBirr, Dashen Bank) appear in banking-domain instances [\[DATASET-D3, D4, D5, D34\]](#). This is materially better than the registry feared. However, several deployment-critical content gaps remain: (a) Ethiopian Birr (ETB) was absent as a CURRENCY/MONEY slot value in any sampled Amharic example despite Naira/Rand appearing in other tracks [\[DATASET-D5, D19, D22 vs Amharic absence\]](#); (b) most Amharic DATE slots used relative expressions rather than named Ge'ez months (only 2/9) [\[DATASET-D15, D16, D28\]](#); (c) Ethiopian time convention (6-hour offset) is unaddressed in TIME annotations [\[DATASET-D11, D23\]](#); (d) no code-switched Amharic–Oromo / Amharic–Tigrinya utterances appear, despite ~70% L2 Amharic users in the deployment population [\[DATASET-D1–D18 monolingual\]](#). IC was marked HIGH priority in elicitation, so these residual gaps weigh heavily.

ID	Evidence
D1	ከድረዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድረዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም እምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው? / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ? / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቱ ላይ ... / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ? / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D7	ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንገረኝ? / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D9	raw: "ለቀጣይ የጉብኝት ጉዞ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding
D10	raw: "ከቴሌብር አካውንቱ ገንዘብ ለማን ለማን እንደላኩ ዝርዝሩን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance
D11	ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"
D12	Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyya irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline
D13	Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar

D14	Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo" / target: "CALENDAR_EVENT: sirna cidhaa \$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ::" / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ::" / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month
D17	በቶቶት ለቃለአብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶት \$\$ PERSONAL_NAME: ቃለአብ — Restaurant "ቶቶት" (Totot) — Ethiopian cultural reference; personal name "ቃለአብ" is distinctly Ethiopian
D18	የቦሎ ማሳደሻ ስንት ነው" / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)
D19	Naira nawa gare ni ... na bankin Gt?" / target: "CURRENCY: Naira \$\$ BANK_NAME: Gt — Nigerian Naira currency annotated in Hausa — confirms per-language currency grounding
D22	Ndinganayo imali engange R700 kwiakhawunti yam yaseStandard Bank?" / target: "MONEY: R700 \$\$ BANK_NAME: yaseStandard Bank — South African Rand (R700) and Standard Bank in Xhosa
D23	"ነገ ጠዋት 12 ሰአት..." (no span annotation) — "12 ሰአት" (12 o'clock) unannotated — "ጠዋት" (morning) absent from target; Ethiopian time convention unclear
D28	በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም::" / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month
D34	raw: "እባካችሁ የዳሽን ባንክ ሂሳቤን ታረጋግጡልኛለችሁ?" / text: "can you please check my dasion bank account" — "Dashen Bank" (major Ethiopian bank) — Amharic source is culturally grounded but English transliteration is lossy ("dasion")

Strengths

- Ethiopian calendar months (Meskerem, Hamle) confirmed as valid DATE slot annotations in travel-domain Amharic instances [DATASET-D1, D2, D6]
- Major Ethiopian intercity cities (Dire Dawa, Bahir Dar, Hawassa, Arba Minch) appear as CITY_OR_PROVINCE slots [DATASET-D2, D9]
- Ground-up cultural elicitation methodology rather than translation [Q6, Q13, Q30]
- Ethiopia-specific banks and TeleBirr appear as natively grounded entities [DATASET-D3, D4, D5]

ID	Evidence
Q6	"The data construction process starts with providing an annotator with sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, and they are to come up with culturally-relevant similar sentences and relevant slot entities (see Figure 1)." (p.1)
Q13	"However, these efforts face two key challenges: (1) the translationese effect, which makes utterances sound less natural in the target languages (Vanmassenhove et al., 2021; Bizzoni et al., 2020), and (2) the creation of utterances that are less culturally relevant." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ:: — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$ CITY_OR_PROVINCE: ባህርዳር \$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው" / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረኝ ንግድን አለ" / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቱ ላይ ..." / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D9	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — deployment demands Amharic city names/abbreviations, Ethiopian calendar dates, Birr fares, and operator-specific terminology, confirmed by elicitation Q2 (HIGH priority) and YAML calendar/toponym sections.

IC-2: Partially. Cultural grounding is confirmed for banking-domain Amharic [DATASET-D3, D4, D5] and partially for travel (Ethiopian cities + Meskerem present) [DATASET-D1, D2, D9]. The grounding methodology is documented [Q30, Q40, Q41].

IC-3: Western-content leakage was not observed in the Amharic sample (though present in some other tracks, e.g., Hausa Taco Bell [DATASET-D31]). Air-travel-to-international-destinations framing dominates the Amharic travel domain [DATASET-D7, D8].

IC-4: Native Amharic speakers annotated [Q7], but regional/L2 demographics undisclosed and code-switched utterances absent from sample [DATASET-D1–D18]; full re-annotation by Ethiopian regional pool would be advisable.

IC-5: Documented residual content gaps: (a) Birr/ETB absent from sampled Amharic CURRENCY/MONEY slots; (b) Ethiopian calendar months sparse vs. relative dates [DATASET-D15, D16, D28]; (c) Ethiopian clock convention undocumented; (d) no code-switched utterances [DATASET-D1–D18].

ID	Evidence
Q2	"Intent detection and slot-filling are crucial components of the natural language understanding module in task-oriented dialogue systems (Hemphill et al., 1990; Coucke et al., 2018; Gupta et al., 2018)." (p.1)
Q6	"The data construction process starts with providing an annotator with sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, and they are to come up with culturally-relevant similar sentences and relevant slot entities (see Figure 1)." (p.1)
Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)
Q40	"A Xhosa annotator was asked to generate another utterance belonging to the same intent type but capturing the South African context where the language is spoken." (p.3)
Q41	"Thus, the annotator used the R200 as "money" with currency Rand (R), and more familiar South African banks such as "FNB" and "Absa" for "bank name" slot." (p.3)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$ CITY_OR_PROVINCE: ባህርዳር \$\$ DATE: መስከረም እምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው" / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ" / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቱ ላይ ..." / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar

D7	ወደ ቱርክ በፍጥነት የምደርስበት የበረራ አማራጭ ንገረኝ። / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D9	raw: "ለቀጣይ የጉብኝት ጉዞ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding
D10	raw: "ከቴሌብር ብር አካባቢ ገንዘብ ለማን እንደላኩ ዝርዝርን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance
D11	ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"
D12	Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyaa irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline
D13	Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar
D14	Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo / target: "CALENDAR_EVENT: sirna cidhaa \$\$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ። / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ትየታ መኪና መከራየት እፈልጋለሁ። / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month
D17	በቶቶ ለቃለኡብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶ \$\$\$ PERSONAL_NAME: ቃለኡብ — Restaurant "ቶቶ" (Totot) — Ethiopian cultural reference; personal name "ቃለኡብ" is distinctly Ethiopian
D18	የቦሎ ማሳደሻ ስንት ነው / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)
D28	በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም። / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month
D31	Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma / target: "RESTAURANT_NAME: taco bell — "Taco Bell" — US fast-food chain, Western entity in Hausa utterance

Information Gaps

- Birr coverage in the 99.4% of Amharic data not sampled
- Frequency of named Ethiopian calendar months vs. relative dates in full travel split
- Whether any code-switched Amharic utterances exist in unsampled portions

Requires Expert Verification

- Native Amharic-speaking expert review of Ethiopian time convention handling in TIME slots across the full dataset

Remediation

Gap 1: Ethiopian Birr (ETB) absent as CURRENCY/MONEY slot in sampled Amharic; most DATE slots use relative expressions rather than named Ge'ez months; Ethiopian clock convention undocumented; no code-switched utterances

Recommendation: Construct a 'ETH-Travel-Slot Stress Set' of Amharic utterances densely populated with (a) Birr fare amounts (VIP/economy tiers), (b) Ge'ez calendar month names across all 13 months including Pagume, (c) Ethiopian-clock TIME values with explicit annotator convention, (d) code-switched Amharic–Oromo and Amharic–Tigrinya tokens for cities and booking terms; evaluate alongside INJONGO Amharic test.

Gap 2: Annotation gaps observed empirically (empty spans on temporally explicit utterances)

Recommendation: Run a targeted audit pass over Amharic train/dev/test for empty-span utterances containing temporal lexemes ('ሰአት', 'ጠዋት', 'ነገ', Ge'ez month names) and patch with corrected annotations before using F1 as a model-selection metric.

Input Form (IF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

INJONGO is a text-only benchmark [Q38, Q43] released under CC BY 4.0 [Q23], with Amharic encoded in Ge'ez/Ethiopic script. The deployment is also a text-based smartphone app, with primary input in Ge'ez script. Dataset inspection confirms all sampled Amharic examples use proper Ge'ez script with no romanization or mixed-script anomalies [DATASET-D1, D5]. There is full modality and script alignment. Elicitation marked IF as LOWER priority, consistent with no concrete violations.

ID	Evidence
Q23	"Dataset is released under CC BY 4.0 license." (p.2)
Q38	"Then, the dataset construction follows two stages: (1) Utterance elicitation in an African language and (2) Slot-filling annotation of the generated utterance." (p.3)
Q43	"The annotation followed the named entity recognition setup on LabelStudio platform." (p.4)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D5	በቴሌብር የብድር አካውንት ላይ ... / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform

Strengths

- Text modality fully matches deployment [Q38]
- Ge'ez script used consistently in all sampled Amharic examples with no anomalies [DATASET-D1, D5]
- CC BY 4.0 license permits adaptation [Q23]

ID	Evidence
Q23	"Dataset is released under CC BY 4.0 license." (p.2)
Q38	"Then, the dataset construction follows two stages: (1) Utterance elicitation in an African language and (2) Slot-filling annotation of the generated utterance." (p.3)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D5	በቴሌብር የብድር አካውንት ላይ ... / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

- IF-1:** Both source and target are plain text in Ge'ez script for Amharic [Q38, DATASET-D1].
- IF-2:** Smartphone IMEs support Ge'ez input on major Android distributions (qualitative web evidence). National 4G coverage of ~33% [WEB-6] affects connectivity but not signal form.
- IF-3:** No domain-specific form differences relevant — both contexts are typed conversational text.
- IF-4:** No form mismatches identified for the Amharic deployment.

ID	Evidence
Q38	"Then, the dataset construction follows two stages: (1) Utterance elicitation in an African language and (2) Slot-filling annotation of the generated utterance." (p.3)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
WEB-6	GSMA Ethiopia: 3G covers ~98% of population, 4G ~33% — connectivity context for smartphone app deployment (https://www.gsma.com/about-us/regions/sub-saharan-africa/wp-content/uploads/2024/10/GSMA_Ethiopia-Report_Oct-2024_v2-1.pdf)

Information Gaps

- Behavior on code-switched tokens written in Latin script (e.g., Oromo lexical insertions) is undocumented

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The 23-slot final ontology [Q118, Q119, Q144] includes structural fits for deployment needs (CITY_OR_PROVINCE, MONEY, CURRENCY, DATE, TIME, BANK_NAME), and merging of low-frequency types is principled [Q51, Q52, Q55]. However, several decision-rule mismatches arise for the Ethiopian deployment: (a) DATE/TIME boundary rules are documented [Q153, Q154] using Gregorian conventions only — no accommodation for Ethiopian calendar (13 months, Pagume intercalary) or 6-hour offset clock; (b) the 40-intent inventory excludes intercity bus, seat-class, fare-tier, and rebooking categories required for Sky Bus/Selam Bus/Ride [Q39, DATASET-D7, D8]; (c) dataset inspection shows Amharic DATE annotations dominated by relative expressions, suggesting the operative decision rule for Ethiopian month names is undertested [DATASET-D15, D16, D28]. These are structural-validity gaps. A redeeming factor: Ethiopian month names did appear as DATE slots [DATASET-D2, D6], so the ontology can host them in principle.

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g. " (p.3)
Q51	"We decided to exclude slot entities appearing less than 500 times across all languages (after MUSIC_GENRE in the figure)." (p.4)
Q52	"Consequently, nine infrequent slots from NATIONALITY through PLUG_TYPE were eliminated." (p.4)
Q55	"This merging process resulted in a reduction from 34 to 23 slot types." (p.5)
Q118	"The "Original Slot Type" are used during the dataset annotation phase, which contained 34 slot types." (p.13)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)
Q144	"# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
D2	target: "CITY_OR_PROVINCE: ድራዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D6	ከሃምሌ 4 እስከ 6 የሰዓት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D7	ወደ ቱርክ በፍጥነት የምድረስበት የበረራ አማራጭ ንገረኝ።" / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል።" / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ።" / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መክራየት እፈልጋለሁ።" / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month

D28	በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም።" / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month
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Strengths

- 23 slot types including travel-relevant CITY_OR_PROVINCE, MONEY, CURRENCY, DATE, TIME [Q144]
- Detailed boundary rules for ambiguous cases (TIME vs DATE, BANK_NAME word inclusion) [Q152, Q153, Q154]
- Ontology can in principle host Ge'ez month names — confirmed empirically [DATASET-D2, D6]
- Merging process reduced ambiguity from 34 to 23 types [Q55, Q119]

ID	Evidence
Q55	"This merging process resulted in a reduction from 34 to 23 slot types." (p.5)
Q119	"After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)
Q144	"# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19)
Q152	"When annotating Bank names, you do not need to include "bank" unless it is attached to the bank name, like seen above, with Ecobank." (p.21)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D6	ከሃምሌ 4 እስከ 6 ያሉት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Most slot types are regionally relevant (cities, money, banks). However, the intent ontology lacks bus-operator-specific categories required for the deployment [Q39, DATASET-D30].

OO-2: Missing: intercity bus intents, seat-class/fare-tier intents, rebooking-on-disruption intent. DATE rule does not encode Ethiopian calendar month set; TIME rule does not encode Ethiopian clock convention [Q153, Q154].

OO-3: DATE/TIME annotation conventions are implicitly Gregorian/international as documented [Q153, Q154]; this encodes a non-Ethiopian temporal convention as default, with no documented annotator instruction for Ge'ez calendar or 6-hour-offset clock.

OO-4: Stakeholder-driven taxonomy redesign is recommended for the intent layer (add bus-specific intents) and for the DATE/TIME decision rules (specify Ge'ez calendar handling and Ethiopian clock disambiguation).

OO-5: Documented issues: missing bus intents harm content validity; Gregorian-only DATE/TIME decision rules harm structural validity for Ethiopian temporal expressions.

ID	Evidence
Q39	“The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g." (p.3)
Q144	“# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME” (p.19)
Q153	“Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples.” (p.21)
Q154	“Anything that is more than one day must be annotated as DATE and not time, as seen above” (p.21)
D30	no intercity corridors, no seat class, no bus operator, no fare tier present — All Amharic travel utterances use air travel framing

Information Gaps

- Whether the annotation guide (full Appendix) includes any Ge'ez calendar examples beyond the verbatim quotes available

Requires Expert Verification

- Whether 'CALENDAR_EVENT' would accommodate Timkat/Meskel as labels — taxonomy intent unclear from documentation

Remediation

Gap 1: DATE/TIME annotation rules implicitly Gregorian/international; no documented decision rule for Ge'ez calendar or Ethiopian 6-hour-offset clock; no intent categories for bus-specific flows

Recommendation: Author a Ge'ez-calendar and Ethiopian-clock annotation addendum to the INJONGO guide, specifying acceptable surface forms for each Ge'ez month, intercalary Pagume handling, and disambiguation rules for ambiguous TIME expressions; add bus-specific intents to the operative ontology for downstream evaluation.

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation methodology is sound on its face: ≥ 3 native annotators per utterance with majority voting [Q42, Q45], Fleiss' Kappa improving to 0.912–1.00 after a two-round review [Q47, Q48], with Amharic specifically reaching $\kappa=0.933$ post-review (initial 0.836). However, several documented and empirical gaps reduce convergent validity for this deployment: (a) annotator regional/L2 demographics within Ethiopia are undisclosed [Q44, gap-4 NOT FOUND]; (b) the actual user population includes L2 Amharic speakers from Oromia, Tigray, and SNNP regions who code-switch — a population mismatch flagged HIGH in elicitation; (c) dataset inspection reveals concrete annotation gaps (empty spans on utterances containing clear DATE/TIME content [DATASET-D11, D33], inconsistent treatment of '12 ሰዓት' across alarm utterances [DATASET-D11 vs D23], misannotated PERSONAL_NAME in Oromo [DATASET-D29]); (d) no code-switched utterances at all in 19 Amharic samples [DATASET-D1–D18]. OC was marked HIGH priority in elicitation, so these residuals weigh heavily.

ID	Evidence
Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q47	"Initial Fleiss' Kappa scores revealed substantial variation across languages, ranging from 0.618 (Zulu) to 0.934 (Shona), indicating significant inter-annotator disagreement." (p.4)
Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም እምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው? / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረኝ ገንዘብ አለ? / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቴ ላይ ... / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ? / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D7	ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንግረኝ? / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D9	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding
D10	raw: "ከቴሌብር አካውንቴ ገንዘብ ለማን እንደላኩ ዝርዝሩን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance

D11	ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"
D12	Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyya irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline
D13	Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa" / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar
D14	Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo" / target: "CALENDAR_EVENT: sirna cidhaa \$\$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ::" / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ::" / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month
D17	በቶቶ ለቃለአብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶ \$\$\$ PERSONAL_NAME: ቃለአብ — Restaurant "ቶቶ" (Totot) — Ethiopian cultural reference; personal name "ቃለአብ" is distinctly Ethiopian
D18	የቦሎ ማሳደሻ ስንት ነው" / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)
D23	"ነገ ጠዋት 12 ሰአት..." (no span annotation) — "12 ሰአት" (12 o'clock) unannotated — "ጠዋት" (morning) absent from target; Ethiopian time convention unclear
D29	Galgala kana caalaan akka dhufuuf dammaksituu guutiif saatii 7tti" / target includes misannotated "PERSONAL_NAME: caalaa — "caalaa" annotated as PERSONAL_NAME but may be part of "Galgala caalaan" (this evening) phrase
D33	የእናቴን እና የአባቴን የልደት ቀን የቀን መቁጠሪያዬ ላይ ስላቀያየርኩት አስተካክለው — "I changed my parents' birthdays in my calendar, please fix it." — no slots annotated despite calendar events

Strengths

- Native-speaker annotation with three annotators and majority voting [Q42, Q45]
- Two-round review process raised inter-annotator agreement substantially [Q47, Q48, Q109]
- Amharic post-review $\kappa=0.933$, second-highest among all initial scores [WEB-17]
- Detailed annotation guide with edge-case rules [Q149, Q156, Q157]

ID	Evidence
Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q47	"Initial Fleiss' Kappa scores revealed substantial variation across languages, ranging from 0.618 (Zulu) to 0.934 (Shona), indicating significant inter-annotator disagreement." (p.4)
Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
Q149	"This section provides the complete annotation guide and instruction for annotators working for labeling all slots types." (p.20)
Q156	"With this entity, only annotate if entity is named explicitly, e.g Name of airport, museum or mall is not and nt just "mall", "airport" etc" (p.24)
Q157	"PS: Do not skip any annotations, if there is nothing to annotate, submit and go to the next one." (p.24)
WEB-17	Amharic post-review $\kappa=0.933$ supports output reliability (https://www.researchgate.net/publication/389056008_INJONGO_A_Multicultural_Intent_Detection_and_Slot-filling_Dataset_for_16_African_Languages)

Checklist

Checklist item definitions:

ID	Assessment Question
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OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially. Native Amharic annotators were used [Q7], but their regional/L2 representativeness for the actual deployment user base (multi-regional, L2-heavy) is undisclosed [Q44].

OC-2: Likely disagreement on (a) code-switched utterances (absent from training entirely), (b) Ethiopian time convention, (c) regional toponyms beyond major cities. Demographic profile of annotators is not published.

OC-3: Documentation specifies 'native speakers' [Q7] and recruitment via a Kenyan logistics company [Q44]. No demographic breakdown by region, urban/rural, age, or L1/L2 status is provided.

OC-4: Re-annotation by a representative pool (including non-Addis L2 speakers and code-switching annotators) is recommended given the deployment's user heterogeneity.

OC-5: Majority voting [Q45] is used. With three annotators and absence of demographic diversity controls, minority regional perspectives may be erased; the review process explicitly resolved disagreement toward consistency [Q53, Q109].

OC-6: Documented issues: undisclosed demographics, observed annotation gaps in Amharic [DATASET-D11, D33], absence of code-switched data — all harming convergent and external validity.

ID	Evidence
Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q53	"Examination of annotator feedback and comparative analysis between unreviewed and reviewed versions indicated that ambiguous slot types significantly impacted annotation quality and introduced unnecessary complexity." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
D11	ነገ ጠዋት 12 ሰዓት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰዓት"
D33	የእናቴን እና የአባቴን የልደት ቀን የቀን መቁጠሪያዬ ላይ ስላቀያየርኩት አስተካክለው — "I changed my parents' birthdays in my calendar, please fix it." — no slots annotated despite calendar events
WEB-3	Amharic L2 lingua-franca status — most learned second language across Ethiopia (https://arxiv.org/html/2603.23654)

Information Gaps

- Annotator demographics within Ethiopia (region of origin, urban/rural, L1/L2 Amharic, age, education)
- Number of annotators recruited for the Amharic split specifically
- Whether any annotation review was performed by Ethiopian regional sub-population

Requires Expert Verification

- Native Amharic speaker review of suspected misannotations and empty-span examples to confirm rate of annotation gaps

Remediation

Gap 1: Annotator regional/L2 demographics within Ethiopia undisclosed; observed annotation gaps (empty spans on utterances with clear DATE/TIME content); no code-switched data ever annotated

Recommendation: Recruit a regionally diverse annotator pool — explicitly including non-Addis annotators from Amhara, Oromia, Tigray, SNNP, and Somali regions, with documented L1/L2 status — and re-annotate a stratified sample (~500 utterances) of the Amharic INJONGO test split plus a new code-switched mini-set; report κ stratified by annotator subgroup.

Score: 4/5 — Minor gaps | **Confidence: high**

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Justification

ID	Evidence
Q19	“Our result shows that fine-tuning baselines could reach an accuracy of 93.7% and F1-score of 85.6 for intent detection and slot-filling tasks respectively.” (p.2)
Q78	“Overall, AfroXLMR-76L achieves the best performance on both ID-SF tasks, with an average accuracy of 93.7% and an F1 score of 85.6%, respectively.” (p.6)
Q84	“Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages.” (p.7)
Q145	“Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return `\$\$` only.” (p.19)
Q146	“Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater” (p.19)
D34	raw: "እባካችሁ የዳሽን ባንክ ሂሳቤን ታረጋግጡልኛለችሁ?" / text: "can you please check my dasion bank account" — "Dashen Bank" (major Ethiopian bank) — Amharic source is culturally grounded but English transliteration is lossy ("dasion")

- Structured intent + slot-tag output format directly matches deployment NLU module requirements [Q145, Q146]
- Accuracy and F1 metrics appropriate for closed-class intent and span-based slot evaluation [Q84]
- Amharic post-review annotation quality high ($\kappa=0.933$) [WEB-17]

ID	Evidence
Q84	“Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages.” (p.7)
Q145	“Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return ``\$` only.” (p.19)
Q146	“Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater” (p.19)
WEB-17	Amharic post-review $\kappa=0.933$ supports output reliability (https://www.researchgate.net/publication/389056008_INJONGO_A_Multicultural_Intent_Detection_and_Slot-filling_Dataset_for_16_African_Languages)

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text-based intent class + slot-tag output exactly matches deployment NLU module needs [Q145, Q146].

OF-2: TTS not relevant — deployment is text-based smartphone app.

OF-3: Smartphone-app users are positively selected for literacy [WEB-4 ~15% smartphone ownership]; text output is appropriate. National literacy ~71% [WEB-1, WEB-2] is lower than the smartphone-using subset, but for the deployment's user segment text output is suitable.

OF-4: Minor: lossy transliteration of Ethiopian entities in the eng config [DATASET-D34] could weaken cross-lingual transfer evaluation but does not affect direct Amharic deployment evaluation.

ID	Evidence
Q84	"Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages." (p.7)
Q145	"Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return ``\$` only." (p.19)
Q146	"Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John \$\$ CITY_OR_PROVINCE Paris \$\$ MONEY 100 \$\$ RESTAURANT_NAME Awater" (p.19)
D34	raw: "እባክህ ደንብህን ለደረሰህ ባንክ ልክህኝ?" / text: "can you please check my dasion bank account" — "Dashen Bank" (major Ethiopian bank) — Amharic source is culturally grounded but English transliteration is lossy ("dasion")
WEB-1	https://statbase.org/data/wld-literacy-rate/
WEB-2	https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ET
WEB-4	Ethiopia smartphone ownership ~15% — text-based output appropriate for app user segment (https://birmetrics.com/ethiopia-narrows-mobile-gender-gap-to-24-but-smartphone-access-for-women-remains-just-6/)

Information Gaps

- Amharic-specific intent accuracy and slot F1 from Table 5 of the paper not directly extracted

Remediation

Gap 1: Lossy transliteration of Ethiopian entities in the eng config used for cross-lingual transfer (e.g., Dashen Bank → 'dasion')

Recommendation: If using INJONGO's English split as a cross-lingual source, normalize transliterations of Ethiopian named entities to standard English spellings (Dashen Bank, Awash Bank, Abay Bank) before training; otherwise prefer in-language SFT.